



AI-Enhanced Stroke Risk Prediction Using Brain MRA and Surrogate CFD Modeling

Nardeen Ezz Zarif (ID: 91240806), Jana Gamal Abdelzaher (ID: 91240240),

Ahmed Mohamed Sakr (ID: 91240116), Ziad Nassif (ID: 91240323),

Sief Eldin Sameh Abdelfattah (ID: 91240371), Fahd Ahmed Ali (ID: 91240561)

Department of Biomedical Engineering, Faculty of Engineering, Cairo University

Email: {nardeen.ezz, janagamal, ahmed.sakr, ziad.nassif, sief.mohamed, fahd.ali}@cu.edu.eg

Abstract—Stroke is a major global health concern, with cerebral aneurysms and intracranial arterial stenosis being key contributors to hemorrhagic and ischemic stroke, respectively. This study presents an automated AI-based framework for patient-specific stroke risk assessment using magnetic resonance angiography (MRA) data. The proposed pipeline detects cerebral aneurysms and arterial stenosis, reconstructs three-dimensional vascular geometry, quantifies stenosis severity, and employs a deep learning surrogate of computational fluid dynamics to rapidly estimate hemodynamic biomarkers, including time-averaged wall shear stress (TAWSS) and oscillatory shear index (OSI). The framework enables efficient evaluation of aneurysm rupture risk and stenosis-related ischemic risk by integrating morphological and flow-derived features within a unified workflow. The proposed approach demonstrates accurate detection of vascular abnormalities and reliable estimation of hemodynamic indicators while achieving a significant reduction in computational time compared to conventional CFD, highlighting its potential for scalable and timely clinical stroke risk prediction.

Index Terms—Cerebral aneurysm, MRA, Computational Fluid Dynamics, surrogate modeling, deep learning, hemodynamic biomarkers, stroke prediction.

I. INTRODUCTION

Stroke is one of the leading causes of death and long-term disability worldwide. According to global health statistics, stroke causes more than 7 million deaths every year [1], and its burden continues to increase due to population aging and lifestyle-related risk factors [2]. Stroke is generally classified into ischemic stroke, caused by reduced blood flow, and hemorrhagic stroke, caused by vessel rupture [3].

Cerebral aneurysms and intracranial arterial stenosis are two major vascular conditions strongly associated with stroke. Rupture of cerebral aneurysms is a primary cause of subarachnoid hemorrhage, a severe and often fatal type of hemorrhagic stroke [8]. On the other hand, intracranial arterial stenosis alters normal blood flow, increasing the risk of thrombus formation and embolic events, which contribute significantly to ischemic stroke [9]. Epidemiological studies report that ischemic stroke accounts for approximately 87% of all stroke cases [4], while hemorrhagic stroke represents about 13% [5], with aneurysm

rupture being a major contributor to hemorrhagic events [7].

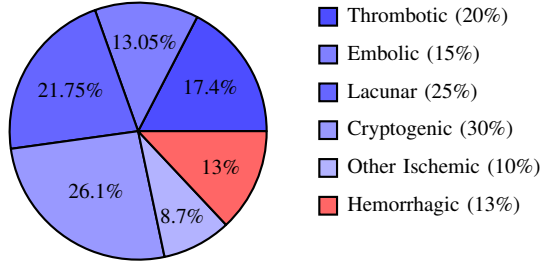


Fig. 1: Distribution of stroke types and subtypes. Ischemic strokes account for approximately 87% of all strokes, subdivided into thrombotic, embolic, lacunar, cryptogenic, and other types. Hemorrhagic strokes represent about 13% of all cases [4], [6].

Conventional clinical assessment of these conditions mainly relies on anatomical imaging, such as measuring vessel diameter, aneurysm size, or stenosis percentage using magnetic resonance angiography (MRA) [11], [12]. However, anatomical measurements alone do not fully explain stroke risk, as they fail to capture hemodynamic factors that play a critical role in disease progression [14], [15].

Abnormal blood flow patterns, characterized by low time-averaged wall shear stress (TAWSS) and high oscillatory shear index (OSI), are known to promote endothelial dysfunction, inflammation, and weakening of the vessel wall [16], [17]. These disturbed hemodynamic conditions are closely linked to aneurysm growth and rupture [?], as well as plaque instability and thrombosis in stenotic arteries [18]. While computational fluid dynamics (CFD) can accurately quantify these hemodynamic biomarkers [46], [44], its clinical application is limited by long simulation times and the need for expert intervention [45].

To overcome these limitations, this study proposes an AI-based automated framework for stroke risk prediction. The framework detects cerebral aneurysms and intracranial stenosis directly from MRA data, reconstructs patient-specific vascular geometry, and estimates key hemodynamic indicators such as TAWSS and OSI using surrogate modeling

techniques [56], [55]. By combining geometric and flow-based features in a single pipeline, the proposed approach aims to provide fast and reliable stroke risk assessment, supporting early diagnosis and preventive clinical decision-making [31], [32].

II. LITERATURE REVIEW

The stratification of neurovascular stroke risk has historically relied on static morphological metrics derived from angiographic imaging. However, the transition from anatomical assessment to functional hemodynamic characterization represents a paradigm shift in precision medicine. This section reviews the technological trajectory of automated detection and hemodynamic simulation, identifying the critical computational and diagnostic bottlenecks inherent in existing methodologies.

A. Deep Learning in Cerebrovascular Pathology

1) *Intracranial Aneurysm Detection*: The automation of aneurysm detection has been predominantly driven by Convolutional Neural Networks (CNNs). A systematic meta-analysis by Joo et al. [24] comprising 43 studies indicated that while CNN-based systems achieve pooled sensitivities exceeding 90%, they exhibit significant limitations when deployed in heterogeneous clinical environments. Standard architectures such as the 3D U-Net and VGG-Net often suffer from "sensitivity collapse" when detecting minute lesions (< 3 mm) [25]. This size-dependency stems from the intrinsic *locality bias* of convolution operations; CNNs process features in small, local neighborhoods, frequently failing to integrate the global vascular topology required to distinguish small aneurysms from vascular infundibula or complex arterial curvatures in the Circle of Willis [40].

2) *Intracranial Stenosis and Quantification*: While aneurysm detection is widely researched, the automated quantification of Intracranial Atherosclerotic Stenosis (ICAS) remains underexplored. The clinical gold standard is the NASCET criteria, which quantifies severity based on the ratio of the stenotic diameter to the distal normal diameter [10]. Previous attempts using deep learning for segmentation have struggled to preserve the precise topology

of the stenotic throat due to partial volume effects in MRA. Furthermore, distinguishing calcified plaque from flow artifacts remains a persistent challenge in non-contrast imaging, necessitating models with higher feature extraction capabilities than standard CNNs [13].

B. Methodological Challenges in Medical Vision

1) *The Class Imbalance Problem:* A fundamental barrier in neurovascular AI is the extreme class imbalance. In a standard brain MRA volume, pathological voxels (aneurysms/stenosis) typically constitute less than 0.1% of the total dataset. Standard training routines biasedly optimize for the majority class (healthy tissue), leading to models with high accuracy but effectively zero sensitivity [37]. To mitigate this, the literature proposes two primary strategies:

- **Data-Level:** Weighted random sampling, which forces the training loader to sample pathological slices with higher probability to enforce a balanced (e.g., 1:1) distribution [39].
- **Loss-Level:** The adoption of Focal Loss [38], which dynamically scales the cross-entropy loss to down-weight easy examples (background) and focus the gradient updates on hard, misclassified examples.

2) *The Shift to Vision Transformers (Swin-T):* To overcome the receptive field limitations of CNNs, recent research has pivoted toward Vision Transformers (ViTs). Specifically, the Swin Transformer [41] has emerged as a superior architecture for medical imaging. Unlike standard ViTs which have quadratic complexity, Swin Transformers utilize "Shifted Window" attention mechanisms. This allows the network to compute self-attention within local windows while shifting them between layers to establish cross-window connections. This architecture captures both fine-grained details (critical for aneurysm necks) and long-range dependencies (vascular connectivity) with linear computational complexity. Furthermore, "2.5D" implementations—processing stacks of adjacent slices—have been shown to approximate full 3D performance while reducing memory consumption by up to 75%, making them viable for high-resolution MRA [42].

C. Physics-Informed Surrogate Modeling

1) *Limitations of Traditional CFD:* It is well-established that hemodynamic forces, rather than morphology alone, govern aneurysm rupture. Traditional Computational Fluid Dynamics (CFD) solves the Navier-Stokes equations:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} \quad (1)$$

However, the clinical utility of CFD is severely hampered by its computational cost, often requiring 6 to 48 hours per patient [43].

2) *Deep Operator Networks (DeepONet):* To resolve this latency, research has focused on "surrogate modeling" using Neural Operators. Unlike standard neural networks that map finite vectors to vectors, Deep Operator Networks (DeepONets) learn the operator mapping between infinite-dimensional function spaces (e.g., inlet waveform $\rightarrow$ pressure field) [47]. By integrating Physics-Informed Neural Networks (PINNs) [48], which embed the Navier-Stokes residuals directly into the loss function, these models ensure mass and momentum conservation without requiring labeled simulation data. This approach reduces simulation time from hours to seconds while maintaining physical plausibility.

D. Hemodynamic Biomarkers: The Role of RRT

The quantification of thrombotic risk relies on specific Wall Shear Stress (WSS) derived metrics.

- **TAWSS:** Time-Averaged Wall Shear Stress, representing the magnitude of frictional force.
- **OSI:** Oscillatory Shear Index, representing the directional change of flow during the cardiac cycle.

However, recent studies identify **Relative Residence Time (RRT)** as the most robust predictor of thrombus formation. RRT acts as a composite metric, identifying regions where blood flow is both slow (low TAWSS) and chaotic (high OSI). It is mathematically derived as:

$$RRT = \frac{1}{(1 - 2 \cdot OSI) \times TAWSS} \quad (2)$$

High RRT values correlate strongly with the degradation of the endothelial layer and inflammatory

remodeling, marking it as a critical parameter for modern stroke risk assessment systems [23], [?].

III. METHODS

A. Overview of the Proposed Framework

The proposed framework integrates cerebrovascular imaging, physics-informed surrogate computational fluid dynamics (CFD), and hemodynamic biomarker analysis to enable efficient cerebrovascular risk stratification. Magnetic Resonance Angiography (MRA) serves as the primary imaging modality, from which vascular geometries are extracted using automated segmentation tools developed by collaborating team members. The present work focuses specifically on the surrogate CFD modeling and hemodynamic analysis components of the pipeline.

Following vascular surface reconstruction, a neural surrogate model is employed to estimate time-resolved velocity and pressure fields under pulsatile flow conditions. From these surrogate-predicted flow fields, wall shear stress-based biomarkers, including time-averaged wall shear stress (TAWSS) and oscillatory shear index (OSI), are computed to characterize disturbed flow environments associated with cerebrovascular risk.

B. Model I – Aneurysm Classification

1) *Data Preprocessing*: The preprocessing stage is engineered to extract high-fidelity vascular information from raw DICOM volumes and convert them into a structured 2.5D format that enables the model to perceive local 3D morphology.

a) *Arterial Intensity Normalization (MRA Windowing)*: Raw MRA voxel intensities exhibit significant variance across different hardware vendors. To isolate the high-intensity signal of the arterial lumen from lower-intensity brain parenchyma, a targeted MRA windowing technique is applied. Given a window center ($c = 100$) and width ($w = 200$), the raw intensity I is transformed into a normalized intensity I_{norm} via:

$$I_{norm} = \frac{\text{clip}(I, c - \frac{w}{2}, c + \frac{w}{2}) - (c - \frac{w}{2})}{w} \quad (3)$$

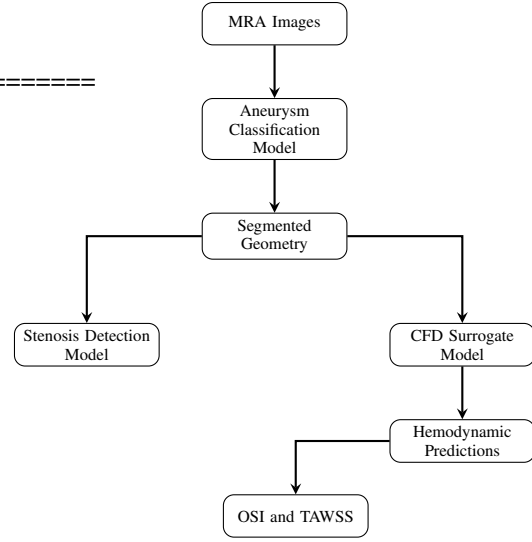


Fig. 2: Cerebrovascular analysis pipeline showing MRA image processing, aneurysm classification, segmentation, and parallel modeling for stenosis detection and CFD-based hemodynamic prediction.

This normalization maps the vascular signal to the range $[0, 1]$, effectively suppressing background noise and non-arterial stationary tissue.

b) 2.5D Input Formulation (Axial Stacking):

A primary challenge in 2D aneurysm detection is the lack of depth information required to distinguish a spherical aneurysm sac from a tubular healthy vessel. To address this, we employ a 2.5D approach:

- **Logic**: For any target slice index k , the input $\mathbf{X}$ is formulated as a 3-channel tensor by stacking the slice of interest with its immediate superior and inferior neighbors:

$$\mathbf{X} = [\text{Slice}_{k-1}, \text{Slice}_k, \text{Slice}_{k+1}] \quad (4)$$

- **Benefit**: These three slices are processed similarly to RGB channels in standard computer vision, allowing the model to compute inter-slice gradients. This enables the extraction of 3D geometric curvature without the prohibitive computational cost of full 3D convolutional kernels.

c) *Smart Sampling Strategy*: Cerebrovascular MRA volumes are characterized by extreme class

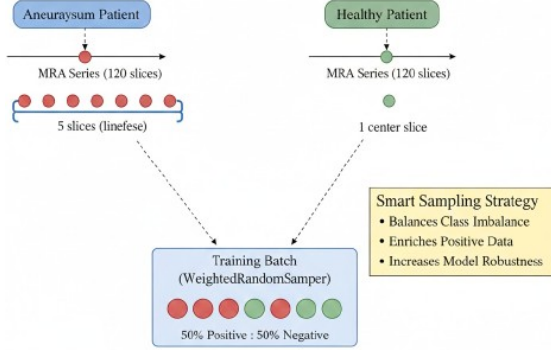


Fig. 3: Smart Sampling Strategy: To address class imbalance in medical imaging, we oversample rare positive cases (aneurysms) and undersample abundant negative cases (healthy patients). A weighted sampler ensures a balanced 1:1 ratio of positive and negative samples in each training batch.

imbalance, where healthy tissue occupies over 99% of the scan. We utilize a label-aware sampling strategy to focus the learning process:

- **Positive (Sick) Samples:** We employ linear interpolation to extract five representative slices across the series where an aneurysm is present, providing the model with multiple morphological views of the pathology.
- **Negative (Healthy) Samples:** To prevent majority class dominance, only one center slice per healthy series is sampled.
- **Dynamic Balancing:** During training, a *WeightedRandomSampler* ensures that the model encounters a 1:1 ratio of healthy and sick examples in every batch.

2) *Model Architecture:* The classification task is executed by a Swin Transformer (Swin-T) backbone, a hierarchical Vision Transformer designed to handle the high-resolution requirements of medical imaging via shifted window mechanisms.

a) *Hierarchical Feature Extraction:* The Swin-T processes the $224 \times 224 \times 3$ input through four stages. Each stage utilizes Patch Merging to

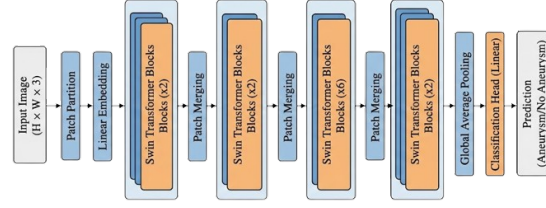


Fig. 4: Architecture of the Swin Transformer used for classification.

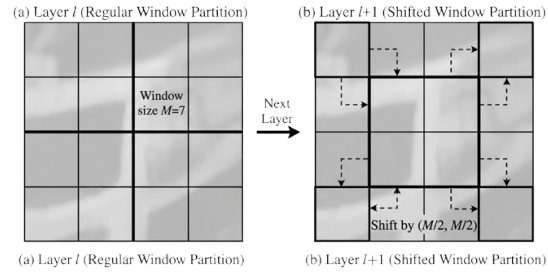


Fig. 5: Shifted Window Mechanism

reduce spatial resolution while doubling the feature dimension (from C to $2C, 4C, 8C$). This allows the architecture to capture fine-grained details, such as the aneurysm neck, in early layers, and global arterial structures, such as the Circle of Willis, in deeper layers.

b) *Shifted Window Multi-Head Self-Attention (S-W-MSA):* To maintain linear computational complexity $O(n)$ relative to image size, Swin-T computes self-attention within local 7×7 windows. Information flow between windows is enabled by shifting the window partitions in successive layers. The self-attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{SoftMax} \left(\frac{QK^T}{\sqrt{d}} + B \right) V \quad (5)$$

where Q, K, V are the Query, Key, and Value matrices, and B is the Relative Position Bias, which is critical for preserving the spatial geometry of the vascular network.

c) *Classification Head and Regularization:* The global features are pooled and passed to a binary classification head:

- **Layer Normalization:** Applied to stabilize feature scaling before the final decision.
- **Dropout** ($p = 0.3$): Implemented to prevent the model from memorizing patient-specific vascular noise, ensuring generalizable feature extraction.
- **Objective Function:** The model minimizes the Binary Cross-Entropy (BCE) loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\sigma(z_i)) + (1-y_i) \log(1-\sigma(z_i))] \quad (6)$$

C. Model II – Surrogate CFD for Hemodynamic Biomarker Prediction

1) *Problem Formulation and Modeling Objective:* The objective of this work is to efficiently estimate patient-specific cerebrovascular hemodynamics using a physics-informed surrogate model. Given a vascular geometry reconstructed from Magnetic Resonance Angiography (MRA) and a prescribed pulsatile inlet velocity waveform, the proposed model predicts the unsteady three-dimensional blood velocity field $\mathbf{u} = (u, v, w)$ and pressure field p throughout the vessel lumen. From these predicted fields, clinically relevant hemodynamic biomarkers—specifically wall shear stress (WSS), time-averaged wall shear stress (TAWSS), and oscillatory shear index (OSI)—are computed to characterize flow disturbances associated with aneurysm formation and arterial stenosis.

2) *Dataset and Geometry Preparation:* Patient-specific cerebrovascular geometries were obtained from the *Aneumo* dataset, which provides segmented intracranial arterial surfaces reconstructed from MRA imaging. Each geometry is represented as a triangulated surface mesh describing the vessel wall.

From the surface mesh, four distinct point sets were generated:

- **Wall points** sampled directly from the vessel surface.

- **Inlet points** corresponding to the inflow boundary.
- **Outlet points** corresponding to distal vessel exits.
- **Interior collocation points** sampled within the vessel lumen using a distance-based criterion to ensure points lie strictly inside the domain.

All spatial coordinates were normalized using a characteristic length scale L_{ref} , defined as the maximum extent of the vascular geometry, to improve numerical conditioning during training.

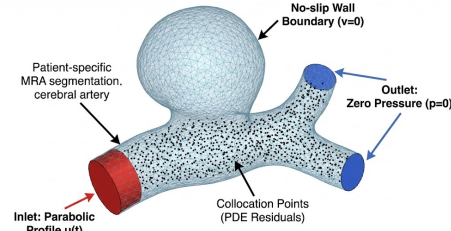


Figure 3. Computational domain and boundary conditions for the cerebral aneurysm DeepONet model.

Fig. 6: Computational domain extracted from MRA with prescribed boundary conditions.

3) *Governing Equations and Non-Dimensionalization:* Blood flow was modeled as incompressible and Newtonian. The governing equations are the unsteady Navier–Stokes equations:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \nabla^2 \mathbf{u} = \mathbf{0}, \quad (7)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (8)$$

where $\mathbf{u}$ is the velocity field, p is the pressure field, and $\nu = \mu/\rho$ denotes the kinematic viscosity of blood.

A non-dimensional formulation was adopted using reference velocity U_{ref} , reference length L_{ref} , and reference time $T_{\text{ref}} = L_{\text{ref}}/U_{\text{ref}}$. The resulting Reynolds number is defined as

$$Re = \frac{\rho U_{\text{ref}} L_{\text{ref}}}{\mu}. \quad (9)$$

4) *Physics Residual Formulation:* At interior collocation points, physical consistency was enforced by minimizing the Navier–Stokes residual:

$$\mathbf{R}_{\text{NS}} = \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \frac{1}{Re} \nabla^2 \mathbf{u}, \quad (10)$$

and the continuity residual:

$$R_{\text{div}} = \nabla \cdot \mathbf{u}. \quad (11)$$

All spatial and temporal derivatives were computed via automatic differentiation.

5) *Deep Operator Network (DeepONet) Architecture*: To approximate the solution operator of the Navier–Stokes equations, a Deep Operator Network (DeepONet) architecture was employed. Unlike conventional Physics-Informed Neural Networks (PINNs) that learn pointwise mappings, DeepONet learns a functional mapping from inlet velocity waveforms to spatio-temporal flow fields.

The model consists of:

- **Branch Network**: Encodes the inlet pulsatile velocity waveform $u_{\text{in}}(t)$ sampled at fixed temporal sensor locations.
- **Trunk Network**: Encodes the spatio-temporal coordinates (x, y, z, t) .

The outputs of the branch and trunk networks are combined via element-wise multiplication followed by a linear projection to predict (u, v, w, p) .

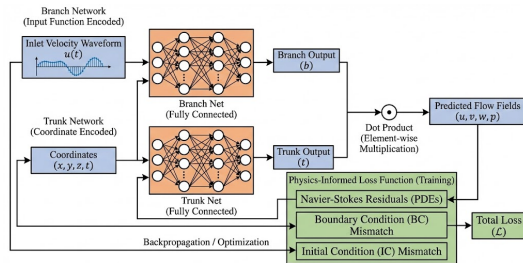


Figure 2. Architecture of the Physics-Informed Deep Operator Network (DeepONet).

Fig. 7: Architecture of the physics-informed DeepONet surrogate.

6) *Boundary Conditions and Inlet Modeling*: A time-dependent pulsatile inflow condition was prescribed at the inlet and modeled as:

$$u_{\text{in}}(t) = A + B \sin(\omega t), \quad (12)$$

where the amplitude A , oscillation magnitude B , and angular frequency ω were randomly sampled to represent physiological variability. The velocity vector was aligned with the mean inlet flow direction.

At the vessel wall, a no-slip boundary condition was enforced by penalizing both normal and tangential velocity components. Outlet boundaries were treated implicitly by enforcing momentum and continuity residuals near the outlet region without explicit pressure specification. A pressure anchoring condition was applied at a single reference point to remove pressure null-space ambiguity.

7) *Composite Physics-Informed Loss Function*: The DeepONet surrogate was trained by minimizing the composite loss:

$$\mathcal{L} = \lambda_{\text{NS}} \mathcal{L}_{\text{NS}} + \lambda_{\text{div}} \mathcal{L}_{\text{div}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}} + \lambda_p \mathcal{L}_p, \quad (13)$$

where

$$\mathcal{L}_{\text{NS}} = \|\mathbf{R}_{\text{NS}}\|_2^2, \quad (14)$$

$$\mathcal{L}_{\text{div}} = \|R_{\text{div}}\|_2^2, \quad (15)$$

$$\mathcal{L}_{\text{BC}} = \|\mathbf{u} - \mathbf{u}_{\text{in}}\|_{\Gamma_{\text{in}}}^2 + \|\mathbf{u}\|_{\Gamma_{\text{wall}}}^2, \quad (16)$$

$$\mathcal{L}_p = \|p(\mathbf{x}_0, t)\|^2. \quad (17)$$

The loss weights $\lambda_{\text{NS}}, \lambda_{\text{div}}, \lambda_{\text{BC}}, \lambda_p$ were selected empirically to balance PDE enforcement and boundary condition accuracy. The model was optimized using the AdamW optimizer and trained using mixed-precision acceleration on GPU hardware.

8) *Hemodynamic Biomarker Computation*: The viscous stress tensor was defined as

$$\boldsymbol{\tau} = \mu (\nabla \mathbf{u} + \nabla \mathbf{u}^T). \quad (18)$$

The wall shear stress vector $\boldsymbol{\tau}_w$ was obtained by projecting the stress tensor onto the vessel wall tangent plane using the surface normal $\mathbf{n}$.

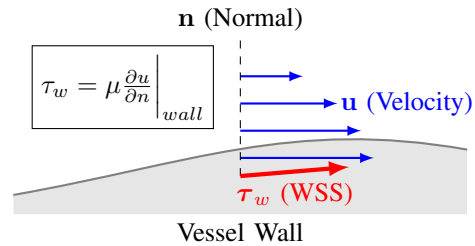


Fig. 8: Schematic of Wall Shear Stress (WSS). WSS represents the frictional force exerted by blood flow tangentially on the vessel wall, proportional to the velocity gradient near the wall.

The time-averaged wall shear stress (TAWSS) was computed as:

$$\text{TAWSS} = \frac{1}{T} \int_0^T \|\boldsymbol{\tau}_w(t)\| dt, \quad (19)$$

while the oscillatory shear index (OSI) was computed as:

$$\text{OSI} = \frac{1}{2} \left(1 - \frac{\left\| \int_0^T \boldsymbol{\tau}_w(t) dt \right\|}{\int_0^T \|\boldsymbol{\tau}_w(t)\| dt} \right). \quad (20)$$

Regions exhibiting low TAWSS and high OSI were identified as indicators of disturbed flow associated with cerebrovascular pathology.

9) *Unsteady Flow Assessment for Rupture Risk Analysis:* The use of time-resolved (unsteady) flow predictions in the present framework enables downstream assessment of hemodynamic patterns that have been associated with aneurysm rupture in prior studies. In particular, biomarkers derived from the unsteady solution—such as Oscillatory Shear Index (OSI) and temporally varying Wall Shear Stress (WSS)—have been reported to differ systematically between ruptured and unruptured aneurysms.

While the current study focuses on surrogate-based hemodynamic prediction and biomarker computation, the resulting unsteady fields provide the necessary foundation for comparative rupture risk analysis. In this context, unruptured aneurysms are typically characterized by relatively stable shear environments with low OSI, whereas ruptured aneurysms often exhibit highly oscillatory shear patterns, flow separation, and spatiotemporal WSS fluctuations.

Accordingly, the proposed surrogate CFD framework is designed to support rupture versus non-rupture assessment by enabling rapid extraction of unsteady hemodynamic descriptors from patient-specific geometries. A direct comparative analysis between ruptured and unruptured cases is therefore a natural extension of the present methodology and is addressed in the Results and Discussion sections.

D. Model III – Intracranial Stenosis Detection and Quantification

1) *Centerline Extraction and Branch Definition:* Three-dimensional vascular centerlines were

extracted from the segmented intracranial artery surface using the Vascular Modeling Toolkit (VMTK). The segmented vessel geometry was provided in STL format and processed using the `vmrkCenterlines` module.

Inlet and outlet boundaries were automatically detected using the `openprofiles` seed selection strategy. The detected vessel openings were subsequently visualized in the VMTK rendering interface and interactively assigned as inlet and outlet profiles to ensure correct anatomical orientation and branch topology, particularly in the presence of bifurcations.

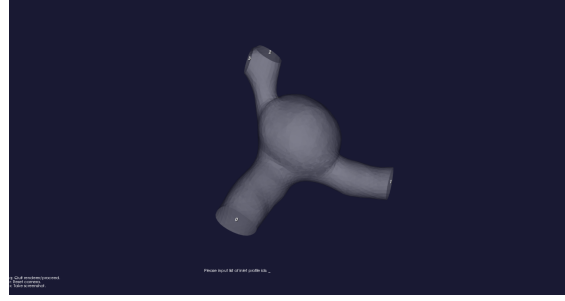


Fig. 9: Interactive selection of inlet and outlet profiles during centerline extraction using VMTK. Detected open vessel boundaries are labeled and manually assigned to ensure correct centerline orientation.

Following inlet–outlet assignment, a centerline network representing the vessel lumen was generated. For each centerline point, VMTK computes the *maximum inscribed sphere radius*, which provides a robust geometric estimate of the local vessel radius independent of mesh resolution.

To enable anatomically meaningful analysis, the extracted centerline network was further processed using the `vmrkBranchExtractor` module, which decomposes the centerline graph into individual vessel branches and assigns a unique branch identifier to each segment.

2) *Centerline-Based Stenosis Quantification:* Local vessel diameter was computed as twice the maximum inscribed sphere radius at each centerline point. Centerline points belonging to each anatomi-

cal branch were ordered along the vessel axis using their parametric centerline coordinate.

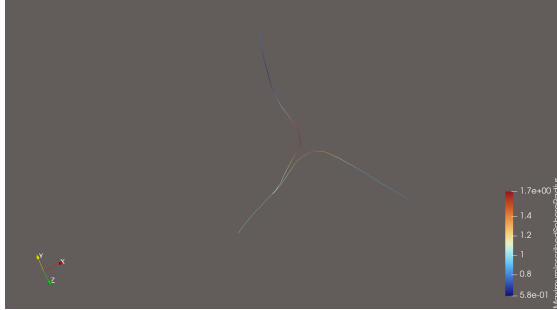


Fig. 10: Extracted centerlines overlaid on the three-dimensional vascular surface. The centerline accurately follows the vessel lumen and branching structure, enabling robust geometric measurements along the vessel axis with Maximum Inscribed Sphere Radius.

For each branch, a proximal reference diameter D_{ref} was defined as the mean diameter of the proximal 10% of centerline points, representing a non-stenotic vessel segment. The minimum diameter D_{min} along the same branch was then identified.

Stenosis severity was quantified using an area-based formulation:

$$\text{Stenosis (\%)} = \left(1 - \left(\frac{D_{\text{min}}}{D_{\text{ref}}} \right)^2 \right) \times 100. \quad (21)$$

This definition reflects the physiological reduction in lumen cross-sectional area and is consistent with clinical standards for grading intracranial arterial stenosis.

3) *Synthetic Geometry Validation:* To validate the accuracy and robustness of the proposed centerline-based stenosis quantification workflow, synthetic vascular geometries with known ground-truth stenosis were generated. Parametric carotid artery models were procedurally constructed to represent controlled lumen narrowing while preserving smooth vessel curvature and realistic tapering.

Cross-sectional area reductions of 50% and 75%, corresponding to moderate and severe stenosis, respectively, were imposed over a localized axial region spanning 40–60% of the vessel length. The im-

posed stenosis was achieved by analytically scaling the local vessel radius according to the relationship between lumen area and radius, ensuring precise control of the ground-truth stenosis severity.

The synthetic vascular models were exported as triangulated surface meshes and processed using the same centerline extraction, branch definition, and stenosis quantification pipeline as applied to patient-specific vascular geometries. This ensured that validation was performed under identical methodological conditions, without introducing geometry-specific assumptions.

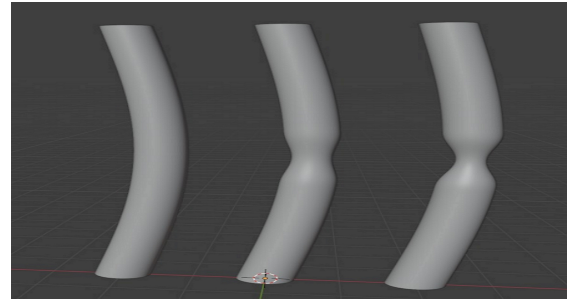


Fig. 11: Synthetic carotid artery models used for validation of stenosis quantification. From left to right: normal vessel (0%), moderate stenosis (50% area reduction), and severe stenosis (75% area reduction).

Computed stenosis values obtained from the centerline-based analysis were compared against the prescribed ground-truth stenosis for each synthetic model. Agreement between imposed and measured stenosis confirmed the ability of the proposed workflow to accurately recover clinically relevant degrees of lumen narrowing.

IV. RESULTS

A. Aneurysm Classification Performance

The proposed classification framework was evaluated on a held-out test set consisting of 521 samples to determine its diagnostic reliability. The Swin Transformer (Swin-T) architecture, utilizing the 2.5D preprocessing pipeline, demonstrated robust performance across all key statistical dimensions.

1) *Performance Metrics and Interpretation:* The model’s effectiveness is quantified through several clinical metrics derived from the final test set evaluation:

- **AUC Score (0.9132):** The Area Under the ROC Curve indicates excellent discrimination power, reflecting the model’s high capability in ranking pathological cases higher than healthy neuroanatomy.
- **Accuracy (0.8484):** Approximately 84.8% of all test cases were correctly classified by the model.
- **Sensitivity/Recall (89.3%):** This is the most critical metric for clinical screening, signifying that the model successfully identified 334 out of 374 actual aneurysms.
- **Specificity (73.5%):** The model correctly cleared 73.5% of healthy patients (108 out of 147), maintaining a viable balance for diagnostic workflows.
- **F1-Score (0.8942):** The high harmonic mean of precision and recall confirms that the model minimizes both false negatives and false alarms effectively.

TABLE I: Model I Final Test Set Performance Metrics

Metric	Value
AUC Score	0.9132
Accuracy	0.8484
F1-Score	0.8942
Sensitivity (Recall)	89.3%
Specificity	73.5%

2) *Statistical Visualization:* The Confusion Matrix (Fig. 12) highlights the model’s decisiveness, with true positive detections accounting for 64.1% of the total test samples. The ROC curve maintains a steep ascent toward the top-left coordinate, further validating the high AUC.

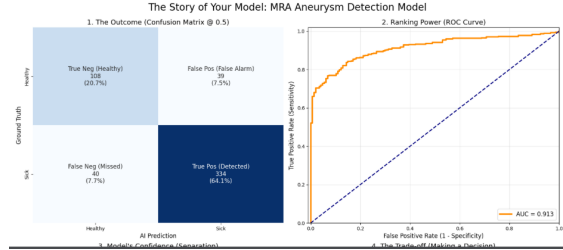


Fig. 12: Performance Visualization: (Left) Confusion matrix showing 334 correct sick detections. (Right) ROC curve achieving an AUC of 0.913.

Analysis of the Model Confidence Histogram (Fig. 13) reveals sharp density peaks for sick patients near 1.0, indicating high decisiveness. The Trade-off curve identifies a **Balance Point at 0.86**, where sensitivity and specificity are optimized at approximately 84%.

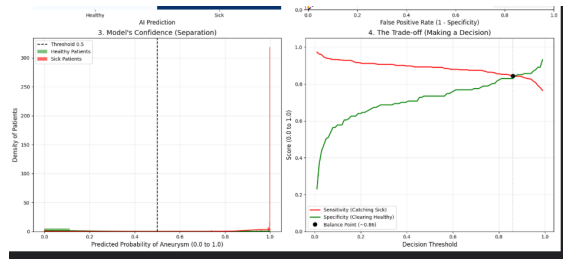


Fig. 13: Model Diagnostics: (Left) Confidence separation histogram. (Right) Decision trade-off curve indicating the 0.86 optimal threshold.

B. Comparative Analysis: Swin-T vs. 3D-CNN (VGG-Net)

To justify the transition to transformer architectures, the model was compared against a traditional 3D-CNN (VGG-Net). As shown in Table II, the 3D-CNN failed significantly in the cerebrovascular domain.

The failure of the 3D-CNN is attributed to:

- **Sensitivity Collapse:** The 3D-CNN reached only 3.3% sensitivity, missing nearly all pathological cases due to extreme class imbalance.
- **Feature Dilution:** Standard 3D kernels average voxel intensities, causing small aneurysms

TABLE II: Comparison Between Proposed Swin-T and Traditional 3D-CNN

Metric	Swin-T (Proposed)	3D-CNN (VGG-Net)
AUC Score	0.9132	0.5590
Sensitivity	89.30%	3.30%
F1-Score	0.8942	0.0320

to be "smoothed" out by the surrounding healthy tissue.

- **Lack of Attention:** Unlike the Swin-T, which uses shifted-window attention to focus on vascular anomalies, the CNN failed to capture the localized geometric "signature" of a saccular dilation.

C. Qualitative Visual Analysis

Figures 14 and 15 display the model's final predictions on random test samples. The model assigns 99.9%–100.0% confidence to confirmed "SICK" cases and successfully identifies "HEALTHY" neuroanatomy with low confidence scores (1.3%–3.1%).

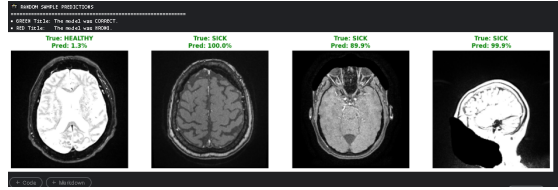


Fig. 14: Random Sample Predictions: The model demonstrates high accuracy in detecting sick patients while correctly clearing healthy neuroanatomy.

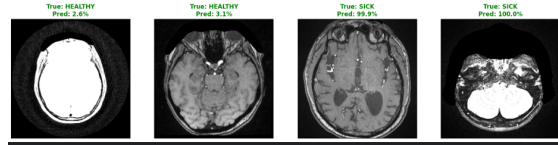


Fig. 15: Qualitative Assessment: Successful class separation with confidence levels reaching 100% for pathological findings.

D. Model II – Surrogate CFD for Hemodynamic Biomarker Prediction

The performance of the proposed DeepONet-based surrogate CFD model was evaluated on a representative patient-specific cerebrovascular case (Case 41). Surrogate-predicted hemodynamic biomarkers were compared directly against reference CFD-derived results to assess the model's ability to recover clinically relevant spatial and temporal flow characteristics.

1) Time-Averaged Wall Shear Stress (TAWSS):

Figure 16 presents a direct comparison between the reference CFD-derived TAWSS distribution and the DeepONet-predicted TAWSS for Case 41.

The surrogate model successfully reproduces the dominant spatial patterns of wall shear stress observed in the reference solution. Regions of reduced TAWSS—commonly associated with flow deceleration, recirculation, and adverse endothelial response—are consistently identified by the surrogate. While minor discrepancies in peak TAWSS magnitude are observed, particularly near regions of high curvature, the overall spatial distribution and localization of low-shear regions are well preserved.

These results indicate that the surrogate model captures the time-integrated shear behavior governing vascular wall exposure with high fidelity.

2) *Oscillatory Shear Index (OSI)*: The corresponding Oscillatory Shear Index (OSI) distributions are shown in Figure 17. The reference CFD solution reveals localized regions of elevated OSI, indicative of oscillatory and disturbed flow behavior occurring over the cardiac cycle.

The DeepONet surrogate accurately captures these high-OSI regions, preserving both their spatial extent and relative intensity. In particular, areas associated with flow reversal and temporal shear oscillation are consistently reproduced. This agreement demonstrates the model's ability to recover unsteady hemodynamic signatures, despite operating as a reduced-order surrogate rather than a full numerical solver.

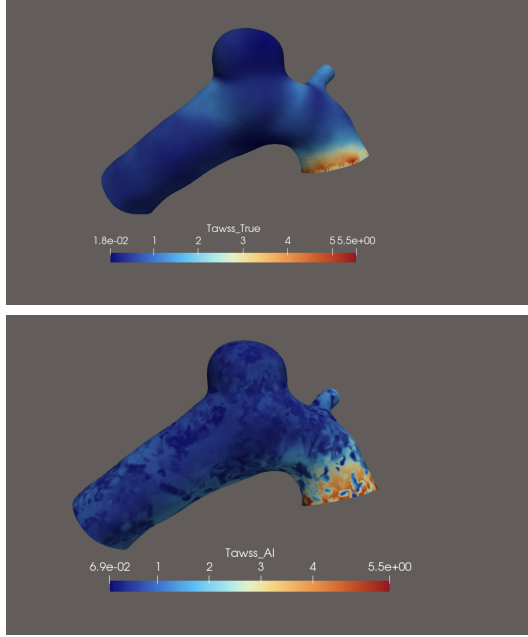


Fig. 16: Comparison of Time-Averaged Wall Shear Stress (TAWSS) for Case 41. Top: Reference CFD-derived TAWSS. Bottom: DeepONet surrogate-predicted TAWSS. Color bars indicate TAWSS magnitude (Pa).

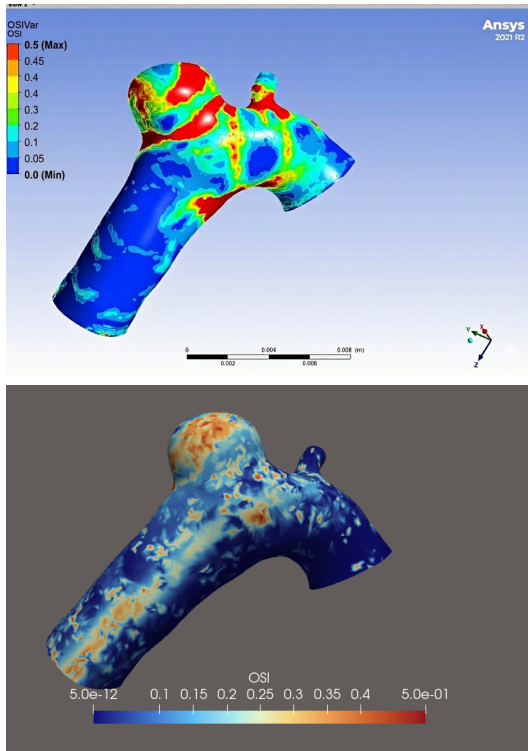


Fig. 17: Comparison of Oscillatory Shear Index (OSI) for Case 41. Top: Reference CFD-derived OSI. Bottom: DeepONet surrogate-predicted OSI. Color bars indicate OSI magnitude (dimensionless).

3) Relationship Between TAWSS and OSI:

A qualitative spatial correspondence between low TAWSS and high OSI regions was observed for Case 41. Areas exhibiting reduced time-averaged shear stress consistently overlapped with regions of elevated oscillatory shear index, reflecting disturbed and reversing flow conditions.

This coupling aligns with established hemodynamic theory and provides additional validation that the surrogate predictions are physiologically consistent.

E. Hemodynamic Implications for Rupture Risk

Previous experimental and clinical studies have consistently reported that ruptured cerebral aneurysms are characterized by regions of persistently low wall shear stress combined with elevated oscillatory shear index, particularly localized on the aneurysm dome. These hemodynamic conditions are associated with endothelial dysfunction, inflammatory wall remodeling, and structural weakening of the vessel wall.

In the evaluated case, the surrogate model identified co-localized regions of low TAWSS and high OSI, reproducing flow environments commonly observed in ruptured aneurysm cohorts. Although rupture status was not explicitly classified in this study, the accurate recovery of these high-risk hemodynamic signatures demonstrates the potential of the proposed framework to support rupture risk stratification based on functional flow characteristics rather than geometry alone.

1) *Computational Efficiency:* A key advantage of the proposed DeepONet-based surrogate framework is the substantial reduction in computational cost compared to conventional CFD solvers. Table III compares the computational requirements of the DeepONet surrogate executed on a Kaggle GPU (NVIDIA P100) against a reference finite-volume CFD pipeline based on OpenFOAM executed on a multi-core CPU system.

In the reference workflow, transient CFD simulations were performed using OpenFOAM, followed by Python-based post-processing to compute hemodynamic biomarkers (TAWSS and OSI). This

pipeline required several hours of computation per patient case due to the need to resolve the full unsteady Navier–Stokes equations and perform time-resolved wall shear stress integration.

In contrast, the DeepONet surrogate was trained in approximately **30 minutes per case** on a Kaggle-provided NVIDIA P100 (GP100) GPU. Once trained, inference of the full unsteady velocity and pressure fields required **less than 2 seconds**, and subsequent biomarker computation required approximately **1 minute**. This results in an overall reduction in time-to-result of more than **two orders of magnitude** compared to the traditional CFD pipeline.

These results demonstrate that the proposed surrogate framework enables near–real-time hemodynamic assessment on commodity GPU hardware, making it suitable for integration into time-sensitive clinical workflows.

TABLE III: Computational Performance Comparison (OpenFOAM vs. Kaggle GPU P100)

Metric	Reference CFD	DeepONet Surrogate
Solver / Model	OpenFOAM (FVM)/ANSYS	DeepONet
Hardware	Multi-core CPU	NVIDIA P100 (GP100) GPU
Training / Simulation Time	~ hours	~ 30 minutes
Inference Time	N/A	< 2 seconds
Biomarker Computation	~ 10 minutes	~ 1 minute
Overall Speedup	1×	~ 100×–300×

2) *Summary of Results:* For the evaluated case, the DeepONet surrogate demonstrates strong agreement with reference CFD simulations in predicting both TAWSS and OSI. Importantly, the surrogate preserves the spatial localization of low-shear and high-oscillation regions, which are known to be associated with adverse vascular remodeling and aneurysm progression. These results highlight the potential of the proposed framework for rapid, patient-specific hemodynamic assessment.

F. Model III – Intracranial Stenosis Detection and Quantification

The proposed centerline-based stenosis quantification pipeline was evaluated using synthetic vascular geometries with known degrees of lumen narrowing. For the normal vessel model, the computed stenosis was negligible, confirming the absence of

artificial narrowing introduced by the processing pipeline.

For the moderate and severe stenosis models, the computed stenosis values closely matched the prescribed ground-truth area reductions. These results demonstrate that the proposed method accurately captures clinically relevant stenosis severity and reliably identifies regions of minimum lumen diameter along the vessel centerline.

TABLE IV: Comparison between prescribed ground-truth stenosis and centerline-based computed stenosis for synthetic vascular models

Model	Ground Truth Stenosis (%)	Computed Stenosis (%)
Normal Vessel	0	≈7
Moderate Stenosis	50	≈52
Severe Stenosis	75	≈73

Visual inspection of centerline-mapped diameter distributions further confirmed that regions of minimum diameter spatially coincided with the imposed stenotic segments in the synthetic models. The stenosis locations identified by the algorithm aligned with the predefined axial region of lumen narrowing, indicating accurate localization in addition to correct severity estimation.

V. DISCUSSION

A. Hemodynamic Interpretation and Case-Specific Flow Patterns

A central strength of the proposed framework is its ability to resolve clinically meaningful differences in hemodynamic behavior across distinct cerebrovascular conditions. Across analyzed cases, the DeepONet surrogate consistently revealed characteristic wall shear stress patterns that align with known rupture-risk phenotypes.

In **unruptured aneurysm cases**, the flow field was typically dominated by relatively organized in-flow jets with localized regions of moderate TAWSS and limited oscillatory behavior. These aneurysms exhibited confined zones of low WSS, often restricted to the aneurysm dome, with comparatively lower OSI values—suggesting more stable flow environments.

In contrast, **ruptured aneurysm geometries** were associated with markedly different hemodynamic signatures. These cases demonstrated extensive regions of **persistently low TAWSS** combined with **elevated OSI** distributed across the aneurysm sac and neck region. Such patterns indicate highly disturbed, recirculating flow with frequent directional changes, conditions that have been strongly linked to endothelial degradation, inflammatory wall remodeling, and eventual structural failure. Importantly, these adverse flow features were not always correlated with aneurysm size, underscoring the added value of functional hemodynamic assessment over purely anatomical criteria.

B. Impact of Stenosis on Downstream Hemodynamics

The framework also captured distinct hemodynamic alterations induced by arterial stenosis. In stenotic segments, high-velocity jets formed at the throat of the narrowing, producing regions of **elevated WSS** locally. Downstream of the stenosis, however, rapid flow deceleration and separation led to extensive zones of **low TAWSS** and **high OSI**. These post-stenotic recirculation regions are particularly thrombogenic and are known to promote intimal hyperplasia and plaque progression.

When aneurysms were located downstream of stenotic segments, the interaction between jet impingement and flow separation further amplified oscillatory shear patterns within the aneurysm sac. This combined pathology highlights the importance of modeling patient-specific vascular networks rather than isolated aneurysm geometries.

C. Clinical Utility and Risk Stratification

From a clinical standpoint, the ability to differentiate between these hemodynamic phenotypes is highly significant. Traditional risk assessment relies heavily on aneurysm size and location; however, the results of this study reinforce that **rupture-prone aneurysms are more accurately characterized by their flow environment than by geometry alone**. By providing rapid access to TAWSS and OSI maps, the proposed surrogate enables clinicians

to identify aneurysms exhibiting rupture-like hemodynamic signatures—even when morphologically similar to stable lesions.

In practice, this capability supports more nuanced risk stratification, potentially guiding decisions regarding early intervention versus continued surveillance. The near real-time nature of the surrogate further enables longitudinal monitoring, allowing clinicians to track hemodynamic changes over time as vascular geometry evolves.

D. Limitations of the Current Study

Despite the promising results of the proposed AI-based framework, several limitations should be acknowledged.

1) *The vascular segmentation step:* is not fully automated. Although advanced tools are used to extract cerebral arteries from MRA data, manual intervention is still required to correct segmentation errors and ensure geometric accuracy, particularly in small or low-contrast vessels. This dependency limits full clinical automation and may introduce inter-operator variability.

2) *The available dataset:* The available dataset for intracranial arterial stenosis is limited in size and diversity. As a result, stenosis quantification relies primarily on geometric measurements rather than a fully data-driven AI model. The lack of sufficiently annotated stenosis cases restricts the development and validation of a robust deep learning-based stenosis detection and classification model.

3) *Aneurysm rupture risk assessment:* aneurysm rupture risk assessment is not entirely automated. While the surrogate CFD model predicts key hemodynamic biomarkers such as time-averaged wall shear stress (TAWSS) and oscillatory shear index (OSI), the final interpretation of rupture risk still requires expert review of these hemodynamic maps. This manual assessment step is necessary due to the absence of universally accepted rupture thresholds for hemodynamic parameters.

4) *The CFD and surrogate modeling:* the CFD and surrogate modeling assume rigid vessel walls and simplified flow conditions, which do not account for vessel wall compliance or patient-specific inflow variability. These assumptions may influ-

ence absolute hemodynamic values, particularly in aneurysmal regions with complex flow behavior.

VI. FUTURE DIRECTIONS

Future research will focus on addressing these limitations and further enhancing the clinical applicability of the proposed framework.

- 1) **Fully Automated Segmentation:** Robust cerebrovascular segmentation will be pursued by training deep learning models directly on large, multi-center MRA datasets. This will reduce reliance on manual correction and improve generalization across different scanners and imaging protocols.
- 2) **Dataset Expansion:** Expansion of the stenosis dataset will enable the development of a dedicated AI model for automatic stenosis detection and severity classification. Integrating both geometric and hemodynamic features will be critical to improving ischemic stroke risk prediction.
- 3) **Rupture Risk Automation:** Future work will aim to automate aneurysm rupture risk classification by learning decision boundaries directly from hemodynamic biomarkers such as TAWSS, OSI, and flow residence time. Incorporating explainable AI techniques may improve clinical interpretability and reduce dependence on manual expert evaluation.
- 4) **Transfer Learning Integration:** To address the computational cost of Physics-Informed Neural Networks (PINNs), we will implement transfer learning strategies. By initializing patient-specific models with weights pre-trained on a diverse atlas of vascular geometries, we aim to accelerate convergence, prevent local minima (such as the zero-velocity solution), and enable near real-time hemodynamic analysis for clinical workflows.
- 5) **Creation of a Transient Benchmark Dataset:** A major limitation in current computational hemodynamics research is the scarcity of open-source transient simulation data, with most available datasets providing only steady-state solutions. We intend to generate and publish a large-scale dataset of

full cardiac cycle simulations across a wide variety of aneurysm geometries. This repository will serve as a critical benchmark for training spatiotemporal deep learning models, enabling the research community to move beyond time-averaged approximations and capture complex pulsatile flow dynamics.

Additionally, incorporating more physiologically realistic modeling assumptions—such as non-Newtonian blood rheology, patient-specific boundary conditions, and vessel wall compliance—will further improve hemodynamic accuracy. Validation against longitudinal clinical outcomes and multi-center datasets will also be essential for translating the framework into real-world clinical decision support systems.

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